

Avoiding sillies

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A motivating quote

To succeed in competition math it is not enough to be right. It is also necessary to not be wrong.

What is a silly?

Definition

A *silly* is an almost-correct solution with a “dumb” mistake.

Sillies are Minnetonka's downfall.

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- Arithmetic mistakes
- Forgetting edge cases
- Misreading the question
- Sign errors

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- Arithmetic mistakes
- Forgetting edge cases
- Misreading the question
- Sign errors
- Listening to a song with a number and writing that instead

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Significant mistakes are not sillies. If there are n problems and you scored m , you didn't silly $n - m$ problems.

- Fundamentally flawed approach
- Failed guess

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- Fundamentally flawed approach
- Failed guess
- Answer sheet spontaneously catches fire
- Accidentally eat chocolate containing alcohol before the contest

The problem

Sillies are easily correctable.

Corollary

Reducing sillies is an easy way to improve.

Example: Me at HMMT 2025

Factors increasing sillies

- Doing problems quickly (Me at ARML tiebreaks 2025)
- Headsolving or solving with low scratch paper
- Sleep deprivation or jet lag
- Strong, distracting emotions, such as excitement, anger, or heartbreak
- Sickness

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- Doing problems under the influence (don't drink and derive)

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- Sickness
- Doing problems under the influence (don't drink and derive)
- Having family history of sillying

Factors increasing ~~decreasing~~ sillies

- Doing problems ~~quickly~~ **slowly**
- ~~Headsolving or solving with low scratch paper~~ **using scratch paper liberally**
- ~~Strong, distracting emotions, such as excitement, anger, or heartbreak~~ **being locked in**

Most important advice

Reasoning must be PAINFULLY explicit. Smart people make stupid mistakes. Write down all reasoning. Do not skip steps in your head!

Slow is smooth, smooth is fast. Silly when barying ISL 2024 G4 turned 20 minutes into 3 hours.

Pointing-and-calling

- In many Asian railways, conductors are required to literally point and call out what they're seeing and doing
- Sounds dumb? Shown to reduce error by 85%
- Being painfully explicit reduces errors

The pointing-and-calling method



Figure: The process of pointing and calling. Image from Wikipedia.

Good and bad scratchwork

Scratchwork isn't only for computation. Work out small examples. Write down reasoning.

Writing **everything** down is pointing-and-calling. If you think 2×3 , don't write 6. Don't skip steps!

Don't use test paper margins (not enough space). You'll write over yourself and problems. You are not Fermat.

Good and bad scratchwork

The following slides include examples of scratch paper.

Think about how their paper is organized and how work is being done.

expected # of cycles

A B C

$$\textcircled{1} 1 \text{ cycle} \rightarrow \frac{3!}{3!} \cdot 1 = \frac{1}{3}$$

$$\textcircled{2} 2 \rightarrow \frac{3!}{3!} \cdot 2 = \frac{1}{1}$$

$$\textcircled{3} 3 \rightarrow \frac{1}{3!} \cdot 3 = \frac{1}{2}$$

$$\left. \begin{array}{l} \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \right\} \frac{11}{6}$$

A B C D

$$\textcircled{1} \rightarrow \frac{3!}{4!} \cdot 1 = \frac{1}{4}$$

$$\textcircled{2} \frac{3+3}{4!} \cdot 2 = \frac{11}{12}$$

$$\textcircled{3} \frac{6}{4!} \cdot 3 = \frac{8}{4}$$

$$\textcircled{4} \frac{1}{4!} \cdot 4 = \frac{1}{6}$$

$$\left. \begin{array}{l} \textcircled{2} \\ \textcircled{3} \\ \textcircled{4} \end{array} \right\} \frac{3+8+9+2}{12} = \frac{25}{12}$$

$$\left(1 - \frac{3!}{4!}\right) c(6) + \frac{3!}{4!} \left(c(6) + 1 \right)$$

$$c(6) + \frac{1}{4}$$

$$\frac{2940}{1781} \cdot \frac{1}{2} = \frac{1470}{1781}$$

$$\frac{1470}{1781} \cdot \frac{1}{2} = \frac{735}{1781}$$

$$\frac{735}{1781} \cdot \frac{1}{2} = \frac{367.5}{1781}$$

$$\frac{367.5}{1781} \cdot \frac{1}{2} = \frac{183.75}{1781}$$

$$\frac{183.75}{1781} \cdot \frac{1}{2} = \frac{91.875}{1781}$$

$$\frac{91.875}{1781} \cdot \frac{1}{2} = \frac{45.9375}{1781}$$

$$\frac{45.9375}{1781} \cdot \frac{1}{2} = \frac{22.96875}{1781}$$

$$\frac{22.96875}{1781} \cdot \frac{1}{2} = \frac{11.484375}{1781}$$

$$\frac{11.484375}{1781} \cdot \frac{1}{2} = \frac{5.7421875}{1781}$$

$$\frac{5.7421875}{1781} \cdot \frac{1}{2} = \frac{2.87109375}{1781}$$

$$\frac{2.87109375}{1781} \cdot \frac{1}{2} = \frac{1.435546875}{1781}$$

$$\frac{1.435546875}{1781} \cdot \frac{1}{2} = \frac{0.7177734375}{1781}$$

$$\frac{0.7177734375}{1781} \cdot \frac{1}{2} = \frac{0.35888671875}{1781}$$

$$\frac{0.35888671875}{1781} \cdot \frac{1}{2} = \frac{0.179443359375}{1781}$$

$$\frac{0.179443359375}{1781} \cdot \frac{1}{2} = \frac{0.0897216796875}{1781}$$

$$\frac{0.0897216796875}{1781} \cdot \frac{1}{2} = \frac{0.04486083984375}{1781}$$

$$\frac{0.04486083984375}{1781} \cdot \frac{1}{2} = \frac{0.022430419921875}{1781}$$

$$\frac{0.022430419921875}{1781} \cdot \frac{1}{2} = \frac{0.0112152099609375}{1781}$$

$$\frac{0.0112152099609375}{1781} \cdot \frac{1}{2} = \frac{0.00560760498046875}{1781}$$

$$\frac{0.00560760498046875}{1781} \cdot \frac{1}{2} = \frac{0.002803802490234375}{1781}$$

$$\frac{0.002803802490234375}{1781} \cdot \frac{1}{2} = \frac{0.0014019012451171875}{1781}$$

$$\frac{0.0014019012451171875}{1781} \cdot \frac{1}{2} = \frac{0.00070095062255859375}{1781}$$

$$\frac{0.00070095062255859375}{1781} \cdot \frac{1}{2} = \frac{0.000350475311279296875}{1781}$$

$$\frac{0.000350475311279296875}{1781} \cdot \frac{1}{2} = \frac{0.0001752376556396484375}{1781}$$

$$\frac{0.0001752376556396484375}{1781} \cdot \frac{1}{2} = \frac{8.761882781982421875}{1781}$$

$$\frac{8.761882781982421875}{1781} \cdot \frac{1}{2} = \frac{4.3809413909912109375}{1781}$$

$$\frac{4.3809413909912109375}{1781} \cdot \frac{1}{2} = \frac{2.19047069549560546875}{1781}$$

$$\frac{2.19047069549560546875}{1781} \cdot \frac{1}{2} = \frac{1.095235347747802734375}{1781}$$

$$\frac{1.095235347747802734375}{1781} \cdot \frac{1}{2} = \frac{0.5476176738739013671875}{1781}$$

$$\frac{0.5476176738739013671875}{1781} \cdot \frac{1}{2} = \frac{0.27380883693695068359375}{1781}$$

$$\frac{0.27380883693695068359375}{1781} \cdot \frac{1}{2} = \frac{0.136904418468475341796875}{1781}$$

$$\frac{0.136904418468475341796875}{1781} \cdot \frac{1}{2} = \frac{0.0684522092342376708984375}{1781}$$

$$\frac{0.0684522092342376708984375}{1781} \cdot \frac{1}{2} = \frac{0.03422610461711883544921875}{1781}$$

$$\frac{0.03422610461711883544921875}{1781} \cdot \frac{1}{2} = \frac{0.017113052308559417724609375}{1781}$$

$$\frac{0.017113052308559417724609375}{1781} \cdot \frac{1}{2} = \frac{0.0085565261542797088623046875}{1781}$$

$$\frac{0.0085565261542797088623046875}{1781} \cdot \frac{1}{2} = \frac{0.00427826307713985443115234375}{1781}$$

$$\frac{0.00427826307713985443115234375}{1781} \cdot \frac{1}{2} = \frac{0.002139131538569927215576171875}{1781}$$

$$\frac{0.002139131538569927215576171875}{1781} \cdot \frac{1}{2} = \frac{0.0010695657692849636077880859375}{1781}$$

$$\frac{0.0010695657692849636077880859375}{1781} \cdot \frac{1}{2} = \frac{0.00053478288464248180389404296875}{1781}$$

$$\frac{0.00053478288464248180389404296875}{1781} \cdot \frac{1}{2} = \frac{0.000267391442321240901947021484375}{1781}$$

$$\frac{0.000267391442321240901947021484375}{1781} \cdot \frac{1}{2} = \frac{0.0001336957211606204509735107421875}{1781}$$

$$\frac{0.0001336957211606204509735107421875}{1781} \cdot \frac{1}{2} = \frac{6.684786058031022548675537109375}{1781}$$

$$\frac{6.684786058031022548675537109375}{1781} \cdot \frac{1}{2} = \frac{3.3423930290155112743377685546875}{1781}$$

$$\frac{3.3423930290155112743377685546875}{1781} \cdot \frac{1}{2} = \frac{1.67119651450775563716888427734375}{1781}$$

$$\frac{1.67119651450775563716888427734375}{1781} \cdot \frac{1}{2} = \frac{0.835598257253877818584442138671875}{1781}$$

$$\frac{0.835598257253877818584442138671875}{1781} \cdot \frac{1}{2} = \frac{0.4177991286269389092922210693359375}{1781}$$

$$\frac{0.4177991286269389092922210693359375}{1781} \cdot \frac{1}{2} = \frac{0.20889956431346945464611053466796875}{1781}$$

$$\frac{0.20889956431346945464611053466796875}{1781} \cdot \frac{1}{2} = \frac{0.104449782156734727323055267333984375}{1781}$$

$$\frac{0.104449782156734727323055267333984375}{1781} \cdot \frac{1}{2} = \frac{0.0522248910783673636615276336669921875}{1781}$$

$$\frac{0.0522248910783673636615276336669921875}{1781} \cdot \frac{1}{2} = \frac{0.02611244553918368183076381683349609375}{1781}$$

$$\frac{0.02611244553918368183076381683349609375}{1781} \cdot \frac{1}{2} = \frac{0.013056222769591840915381908416748046875}{1781}$$

$$\frac{0.013056222769591840915381908416748046875}{1781} \cdot \frac{1}{2} = \frac{0.0065281113847959204576909542083740234375}{1781}$$

$$\frac{0.0065281113847959204576909542083740234375}{1781} \cdot \frac{1}{2} = \frac{0.00326405569239796022884547710418701171875}{1781}$$

$$\frac{0.00326405569239796022884547710418701171875}{1781} \cdot \frac{1}{2} = \frac{0.001632027846198980114422738552093505859375}{1781}$$

$$\frac{0.001632027846198980114422738552093505859375}{1781} \cdot \frac{1}{2} = \frac{0.0008160139230994900572113692760467529296875}{1781}$$

$$\frac{0.0008160139230994900572113692760467529296875}{1781} \cdot \frac{1}{2} = \frac{0.00040800696154974502860568463802337646484375}{1781}$$

$$\frac{0.00040800696154974502860568463802337646484375}{1781} \cdot \frac{1}{2} = \frac{0.000204003480774872514302842319011688232421875}{1781}$$

$$\frac{0.000204003480774872514302842319011688232421875}{1781} \cdot \frac{1}{2} = \frac{0.0001020017403874362571514211595058441162109375}{1781}$$

$$\frac{0.0001020017403874362571514211595058441162109375}{1781} \cdot \frac{1}{2} = \frac{5.100087019371812857557105797529220710546875}{1781}$$

$$\frac{5.100087019371812857557105797529220710546875}{1781} \cdot \frac{1}{2} = \frac{2.5500435096859064287785528987646103552734375}{1781}$$

$$\frac{2.5500435096859064287785528987646103552734375}{1781} \cdot \frac{1}{2} = \frac{1.27502175484295321438927644938230517763671875}{1781}$$

$$\frac{1.27502175484295321438927644938230517763671875}{1781} \cdot \frac{1}{2} = \frac{0.637510877421476607194638224691152588818359375}{1781}$$

$$\frac{0.637510877421476607194638224691152588818359375}{1781} \cdot \frac{1}{2} = \frac{0.3187554387107383035973191123455762944091796875}{1781}$$

$$\frac{0.3187554387107383035973191123455762944091796875}{1781} \cdot \frac{1}{2} = \frac{0.15937771935536915179865955617278814720458984375}{1781}$$

$$\frac{0.15937771935536915179865955617278814720458984375}{1781} \cdot \frac{1}{2} = \frac{0.079688859677684575899329778086394073602294921875}{1781}$$

$$\frac{0.079688859677684575899329778086394073602294921875}{1781} \cdot \frac{1}{2} = \frac{0.0398444298388422879496648890431970368011474609375}{1781}$$

$$\frac{0.0398444298388422879496648890431970368011474609375}{1781} \cdot \frac{1}{2} = \frac{0.01992221491942114397483244452159851840057373046875}{1781}$$

$$\frac{0.01992221491942114397483244452159851840057373046875}{1781} \cdot \frac{1}{2} = \frac{0.009961107459710571987416222260799259200286865234375}{1781}$$

$$\frac{0.009961107459710571987416222260799259200286865234375}{1781} \cdot \frac{1}{2} = \frac{0.0049805537298552859937081111303996296001434326171875}{1781}$$

$$\frac{0.0049805537298552859937081111303996296001434326171875}{1781} \cdot \frac{1}{2} = \frac{0.00249027686492764299685405556519981480007171630859375}{1781}$$

$$\frac{0.00249027686492764299685405556519981480007171630859375}{1781} \cdot \frac{1}{2} = \frac{0.001245138432463821498427027782599907400035858154296875}{1781}$$

$$\frac{0.001245138432463821498427027782599907400035858154296875}{1781} \cdot \frac{1}{2} = \frac{0.0006225692162319107492135138912999537000179290771484375}{1781}$$

$$\frac{0.0006225692162319107492135138912999537000179290771484375}{1781} \cdot \frac{1}{2} = \frac{0.00031128460811595537460675694564997685000896453857421875}{1781}$$

$$\frac{0.00031128460811595537460675694564997685000896453857421875}{1781} \cdot \frac{1}{2} = \frac{0.000155642304057977687303378472824988425004482269287109375}{1781}$$

$$\frac{0.000155642304057977687303378472824988425004482269287109375}{1781} \cdot \frac{1}{2} = \frac{7.78211520289888436516689236412442125022411346145546875}{1781}$$

$$\frac{7.78211520289888436516689236412442125022411346145546875}{1781} \cdot \frac{1}{2} = \frac{3.891057601449442182583446182062210625112056730727734375}{1781}$$

$$\frac{3.891057601449442182583446182062210625112056730727734375}{1781} \cdot \frac{1}{2} = \frac{1.9455288007247210912917230910311053125560283653638671875}{1781}$$

$$\frac{1.9455288007247210912917230910311053125560283653638671875}{1781} \cdot \frac{1}{2} = \frac{0.97276440036236054564586154551555265627801418268193359375}{1781}$$

$$\frac{0.97276440036236054564586154551555265627801418268193359375}{1781} \cdot \frac{1}{2} = \frac{0.486382200181180272822930772757776328139007091340966796875}{1781}$$

$$\frac{0.486382200181180272822930772757776328139007091340966796875}{1781} \cdot \frac{1}{2} = \frac{0.2431911000905901364114653863788881640695035456704833984375}{1781}$$

$$\frac{0.2431911000905901364114653863788881640695035456704833984375}{1781} \cdot \frac{1}{2} = \frac{0.12159555004529506820573269318944408203475177283524169921875}{1781}$$

$$\frac{0.12159555004529506820573269318944408203475177283524169921875}{1781} \cdot \frac{1}{2} = \frac{0.060797775022647534102866346594722041017375886417620849609375}{1781}$$

$$\frac{0.060797775022647534102866346594722041017375886417620849609375}{1781} \cdot \frac{1}{2} = \frac{0.0303988875113237670514331732973610205086879432088104248046875}{1781}$$

$$\frac{0.0303988875113237670514331732973610205086879432088104248046875}{1781} \cdot \frac{1}{2} = \frac{0.01519944375566188352571658664868051025434397160440521240234375}{1781}$$

$$\frac{0.01519944375566188352571658664868051025434397160440521240234375}{1781} \cdot \frac{1}{2} = \frac{0.007599721877830941762858293324340255127172195802202606201171875}{1781}$$

$$\frac{0.007599721877830941762858293324340255127172195802202606201171875}{1781} \cdot \frac{1}{2} = \frac{0.0037998609389154708814291466621701275635860979011013031005859375}{1781}$$

$$\frac{0.0037998609389154708814291466621701275635860979011013031005859375}{1781} \cdot \frac{1}{2} = \frac{0.00189993046945773544071457333108506378179304895055065155029296875}{1781}$$

$$\frac{0.00189993046945773544071457333108506378179304895055065155029296875}{1781} \cdot \frac{1}{2} = \frac{0.000949965234728867720357286665542531890896524475275325775146484375}{1781}$$

$$\frac{0.000949965234728867720357286665542531890896524475275325775146484375}{1781} \cdot \frac{1}{2} = \frac{0.0004749826173644338601786433327712659454482622376376628875732421875}{1781}$$

$$\frac{0.0004749826173644338601786433327712659454482622376376628875732421875}{1781} \cdot \frac{1}{2} = \frac{0.00023749130868221693008932166638563297272413111881883144378662109375}{1781}$$

$$\frac{0.00023749130868221693008932166638563297272413111881883144378662109375}{1781} \cdot \frac{1}{2} = \frac{0.000118745654341108465044660833192816486362065559409415721893310546875}{1781}$$

$$\frac{0.000118745654341108465044660833192816486362065559409415721893310546875}{1781} \cdot \frac{1}{2} = \frac{5.93728271705542325222304165964082431810327777047078609466552734375}{1781}$$

$$\frac{5.93728271705542325222304165964082431810327777047078609466552734375}{1781} \cdot \frac{1}{2} = \frac{2.968641358527711626111520829820412159051638885235393047332763671875}{1781}$$

$$\frac{2.968641358527711626111520829820412159051638885235393047332763671875}{1781} \cdot \frac{1}{2} = \frac{1.4843206792638558130557604149102060795258194426176965236663818359375}{1781}$$

$$\frac{1.4843206792638558130557604149102060795258194426176965236663818359375}{1781} \cdot \frac{1}{2} = \frac{0.74216033963192790652788020745510303976290972130884826183319091796875}{1781}$$

$$\frac{0.74216033963192790652788020745510303976290972130884826183319091796875}{1781} \cdot \frac{1}{2} = \frac{0.371080169815963953263940103727551519881454860654424130916595458984375}{1781}$$

$$\frac{0.371080169815963953263940103727551519881454860654424130916595458984375}{1781} \cdot \frac{1}{2} = \frac{0.1855400849079819766319700518637757599407274303272120654582977294921875}{1781}$$

$$\frac{0.1855400849079819766319700518637757599407274303272120654582977294921875}{1781} \cdot \frac{1}{2} = \frac{0.09277004245399098831598502593188787997036371516360603272914886474609375}{1781}$$

$$\frac{0.09277004245399098831598502593188787997036371516360603272914886474609375}{1781} \cdot \frac{1}{2} = \frac{0.046385021226995494157992512965943939985181857581803016364574432373046875}{1781}$$

$$\frac{0.046385021226995494157992512965943939985181857581803016364574432373046875}{1781} \cdot \frac{1}{2} = \frac{0.0231925106134977470789962564829719699925909287909015081822872161865234375}{1781}$$

$$\frac{0.0231925106134977470789962564829719699925909287909015081822872161865234375}{1781} \cdot \frac{1}{2} = \frac{0.01159625530674887353949812824148598499629546439545075409114360809326171875}{1781}$$

$$\frac{0.01159625530674887353949812824148598499629546439545075409114360809326171875}{1781} \cdot$$

Foot $((a^2+bc)xy/xy-a^2bc, b^2c^2)$

$$\frac{1}{2} (b^2+c^2 + a \frac{(a^2+bc)xy}{xy-a^2bc} - \frac{a^2+bc}{xy-a^2bc})$$

★ If P lies on the ~~external~~ internal bisector:

J lies on the internal bisector

★ If P lies on the internal bisector:

J lies on the external bisector

$$\boxed{p^2 - bc = 0}$$

$$\boxed{j - a \bar{p} \bar{j} = a - p}$$

Foot(J).

$$\frac{(j-a)^2}{(b-a)(c-a)} \in \mathbb{R}$$

$$j-a = a \bar{p} \bar{j} - p$$

$$p-a = a \bar{p} \bar{j} - j$$

$$\frac{(j-a)}{(a-p)} \in i\mathbb{R} \rightarrow \frac{j-a}{(a-p)} = \frac{a \bar{p} \bar{j} - \frac{1}{a}}{\frac{1}{a}}$$

$$2pq \left(\frac{x^2 + yz}{y+z} \right) - (p+q) \left(\frac{x(x+y/(y+z))}{y+z} \right) = \frac{\sum}{(x-p)(x-q)} = \frac{xy + 3yz + zx - y^2}{y+z}$$

$$(y+z) \sum (xy + yz + zx) = 2(x^2 + yz) \cdot yz(x+y+z) - x(x+y/(y+z))(4xyz + (x+y/(y+z))(z+x)) + 2x^2(y+z)(xy + yz + zx)$$

$$2(x-p)(x-q)(xy + yz + zx) = \frac{4}{2} \frac{x}{2}$$

$$x^2(xy + yz + zx) - x(p+q)(xy + yz + zx) + pq(xy + yz + zx) =$$

$$x^2(xy + yz + zx) - \frac{x(x+y/(y+z))(4xyz + (x+y/(y+z))(z+x))}{2} + xyz(x+y+z)$$

$$\frac{4}{2} \frac{x}{2}$$

$$= 2x^2(xy + yz + zx) - x(4xyz + (x+y/(y+z))(z+x)) + 2xyz(x+y+z)$$

$$= 2x^2(xy + yz + zx) - x(x+y/(y+z))(z+x) + xyz(2y+2z-2x)$$

$$= x(2x(xy + yz + zx) - (x+y/(y+z))(z+x) + 2xyz(y+z-x))$$

$$= x(2x^2(y+z) - \sum x^2y + 2yz(y+z) - 2xyz)$$

$$= x(x^2y + x^2z + y^2z + yz^2 - x(y^2 + z^2) - 2xyz)$$

$$= x(x^2(y+z) + yz(y+z) - x(y^2 + z^2 + 2yz))$$

$$= x(y+z)(x^2 + yz - x(y+z))$$

$$= x(y+z)(x-y)(x-z)$$

C1

- if $2xy$ is a perfect square!

$$x = \frac{49^2}{2}$$

20, 21, 22, 23 work

$$x + \log_8(y) = 1024 \rightarrow$$

$$y = 8^{1024-x} \sim 2^{1024-x}$$

- if x is even, then $2xy \sim 2 \times 2^{\frac{1024}{2}}$ is a square

- if x is odd, then $2xy \sim 2 \times 2^{\frac{1024}{2}-1} \sim x$ is a square

$$1^2, 3^2, 5^2, \dots, 31^2 < 1024 \quad 16 \times 22 = 352$$

$$\frac{2^2}{2}, \frac{4^2}{2}, \dots, \frac{49^2}{2} \quad \left(\frac{49^2}{2}\right) > 1024$$

$$46^2 > 2048$$

$N, N+2019$ are palindromes

The last digit goes down by 1



2019

$$\frac{a}{2} \frac{b}{c} \frac{c}{0} \frac{6}{1} \frac{a}{9}$$

(last digit can't be 0)

hence there needs to be an "overflow"

(in particular, we want $2 \rightarrow 1$)

$$90 + n \equiv 1 \pmod{p-1} \text{ oh.}$$

$$p \mid 2^n + 3$$

$$a^n + b$$

$$b^n + a$$

$$\rightarrow a^n - b^n + b - a \equiv$$

$p \mid$ some element of $3^n + 2$

fix some ~~prime~~ large prime p . is it reasonable that ~~can we get~~

$$p \mid (2^n + 3, 3^n + 2)?$$

p divides some element of $\{2^n + 3\}$.

$\frac{1}{p}$ -ish coincidence that the "periods match $\frac{1}{p}$ " p divides some element of $\{3^n + 2\}$

but $\sum \frac{1}{p}$ diverges $\rightarrow$ done.

$$p \mid 2^n + 3 \equiv 0 \pmod{p}$$

$$3 \equiv -2^n \pmod{p}$$

$$2^{p-1} \equiv 1 \pmod{p}$$

$$3^m \equiv (-2^n)^m \equiv -2 \pmod{p}$$

$$mn \equiv 1 \pmod{p-1}$$

$$\cancel{2^{mn}} \quad 2^{mn}$$

m even odd

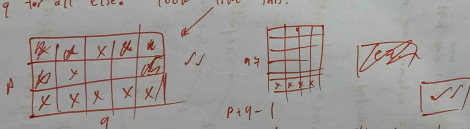
$$(-2^n)^m \equiv 3^m \equiv -2 \pmod{p}$$

$$\equiv 2^{mn} \equiv 2 \pmod{p}$$

- The set must be surjective modulo all but at most one prime.
(Proof. If else, then you take that cross-section, it's ambiguous)
- The graph must be connected.

Answer: $2 \cdot 87 - 1$.

Construction: Pick everything to be 0 mod 97. Then just make the grid mod pq for all else. look like this.



This works — it detects periodicity mod 9 with 40 always.

Bound. 87, 89, and 97, pick one, and



$2 \cdot 87 - 1$ is the bound



$$89 + 87 - 1 = 175$$

$p_1, \dots, p_k$ are the primes < 100 .

we need to find (for all but 1)

$i \neq j$ s.t. $f(i) \neq f(j)$ ~~is not~~ in $\mathbb{Z}(p)$

2×89 works, yet? The bound is not good — we ~~can~~ can make it ambiguous between 89 and 97. Is the answer smaller?

Can we choose $2 \times 89 - 2$?

like
choose 10
such that

(mod 5)

4	X				X
3			X		
2		X			X
1	X			X	
0	X				
	0	1	2	3	4

oh, we are playing "perfectly":

~~Each 2880~~

If we only pick 9, what goes wrong?

(mod 7)

on its own

Bob: ~~From~~ ~~10~~ ~~10~~ element on its own

1			0	
		0		
				2
1				

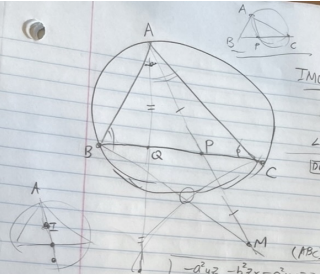
select $2q-1$ squares such that
it is impossible to tell them ~~the~~ s.t.
ambiguity — it two elements are

it two elements in same row, same value

it two elements in same column, same value

→ 99.

[illegible]



Def 1 $u \cdot v \cdot w = 1$
 $(u, v, w) = u\vec{A} \cdot v\vec{B} \cdot w\vec{C}$

IMO 2014 #4
 $\angle PAB = \angle BCA$
 $\angle CAQ = \angle ABC$

Def 2 

Def 3 $(u, v, w) = (u, v, w)$
 $\frac{u}{a} = \frac{v}{b} = \frac{w}{c}$

$I = \left(\frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}} \right)$
 $I = (a : b : c)$

$\vec{P} = \frac{1}{3}(\vec{A} + \vec{B} + \vec{C})$
 $(0 : \frac{1}{3} : \frac{1}{3})$
 $\frac{1}{3}\vec{B} + \frac{1}{3}\vec{C}$

$-a^2yz - b^2zx - c^2xy = 0$

$P = (0 : a^2 - c^2 : c^2) = (0, \frac{a^2 - c^2}{a^2}, \frac{c^2}{a^2})$
 $Q = (0 : b^2 : a^2 - b^2) = (0, \frac{b^2}{a^2}, \frac{a^2 - b^2}{a^2})$
 $\vec{M} = 2\vec{P} - \vec{A} = [2(\frac{c^2}{a^2}\vec{B} + \frac{c^2}{a^2}\vec{C})] - \vec{A}$
 $= (-1, \frac{2c^2 - 2c^2}{a^2}, \frac{2c^2}{a^2}) = (-a^2 : 2c^2 : 2c^2)$
 $\vec{N} = (-a^2 : 2b^2 : 2a^2 - 2b^2)$

$\vec{BN} = \{(-a^2 : \star : 2c^2)\}$
 $2c^2x + a^2z = 0$
 $xz = -a^2z^2$

$\vec{CN} = \{(-a^2 : 2b^2 : \star)\}$
 $(-a^2 : 2b^2 : 2c^2)$

$$\sum a_i = r + \binom{r+1}{3}$$

$$\sum a_i = \binom{r+1}{3} + \binom{r+1}{3} + \binom{2n-r+1}{3}$$

$$\sum_{i=1}^n a_i = r + \binom{r}{3} + \binom{2n-r+1}{2} + \binom{2n-r}{2}$$

$$\sum d_i = \left[2n \binom{r}{2} + \binom{2n-r}{2} \binom{r+1}{2} + r(2n-r)^2 + 2 \binom{2n-r+1}{3} \right]$$

$$\sum d_i - a_i = (2n-r)r + r(2n-r)^2 + 2n \binom{r}{2} + 2 \binom{2n-r+1}{3} - r - \binom{r}{3} - \binom{2n-r+1}{3}$$

$$\frac{d}{dr} : 2n - 2r + 3r^2 - 8rn + 4n^2 + n(2r+1) + \frac{-3(2n-r)^2}{2} - 1 = \frac{3r^2 - 6r + 4n}{2} - \frac{2n-1}{2}$$

$$\frac{3}{2}r^2 + r(-2 - 8n + 4n + 1) + (2n + 4n^2 - 4n) + \frac{1}{3} - 1 - \frac{1}{6} + \frac{2n}{3} + \frac{1}{2}$$

$$\frac{3}{2}r^2 - r(4n+2) + (3n - \frac{1}{2})$$

$$\rightarrow r \approx \frac{4n}{3} - \frac{1}{6}$$

$2 \times 10 \rightarrow 2 \times 50$
 add n ? Prove w/ 10, guaranteed 3.
 must be on an edge?

7 left, 3 markers.
 10 total, 13 markers.
 9 left.

both down!
 cannot distinguish
 up and down.
 $\Rightarrow$ down is always
 one middle.

bad.
 this 9×9 square must also avoid. the one @ and it's also different.

In general n^2 you can do this? want n^2+1 , period n^2 ?
 $n-1$ markers left, n^2 squares $\rightarrow \frac{1}{n} (n^2 - (n-1)) = \frac{1}{n}$ done. ✓ BUT!!

yay!!!

yes, this works. Any number does in 1 stream
 well!

$(j, 9-j)$
 partition positive integers into Fibonacci.
 $\hookrightarrow$ must include 1, 2, 3, 5, 8, 13, ...
 to include 1 and 2, 3.
 normal $\rightarrow$ Fibonacci. $a+b, a+2b, 2a+b, \dots$
 $\hookrightarrow F_{n+1} + F_{n+2}$ general form.

different sequences
 all fine except for 4, 6, 7.
 (!!)

claim: at same point $F_{n+1} + F_{n+2} = F_n$ for any n ?
 can even say that a must be 4 form out, 1, 2, 3, 5, sequence.
 $4, 6, 10, 16, 26, 42, \dots$
 $\rightarrow$ overlap
 could be 1, 4, 6, ...
 this doesn't overlap with the Fibonacci sequence
 by bounding.

what about other sequence? $\rightarrow$ E.g. 3x already doesn't work here.

Handwritten notes on a piece of paper:

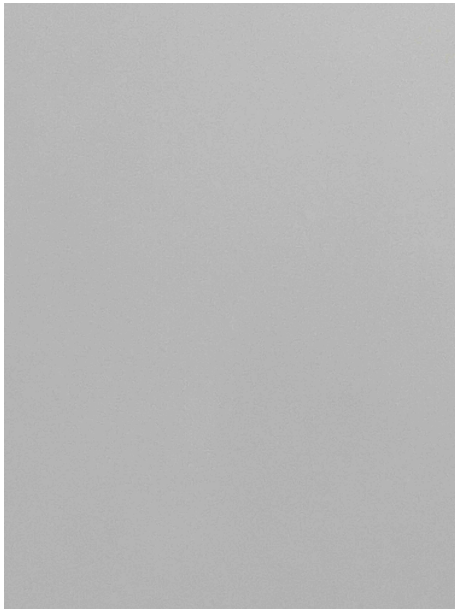
$n=1$ 1

$n=2$ $1 \cdot 4 = 4$
 $1+2 = 3$
 $1+2 = 3$

$n=3$ $\sqrt{5}$
 $4 \cdot 1 + 1 \cdot 2 = 6$

$n=4$ $4 \cdot 1 + 1 \cdot 2 + 1 \cdot 4 = 10$

or this is final.



Practice meet today

We'll do Sample Meet 4 Coyote. Write down reasoning in words. I'll check! If you do not do this, you will face execution.

Good luck!